

SHELDON, IA, USA

WMO: 725495

Lat: 43.208N

Lon: 95.833W

Elev: 1419

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-5.6	-17.1	2.3	-9.5	-11.5	3.1	-4.7	33.4	21.1	30.4	17.6	7.0	330	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	90.9	75.4	88.4	74.2	85.9	73.2	79.9	86.6	77.8	84.8	75.8	82.9	13.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	150.7	83.8	75.3	139.6	82.2	73.1	129.8	80.3	44.7	87.4	42.2	85.3	40.4	81.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.5	25.5	22.7	-21.2	95.7	5.2	3.6	-24.9	98.3	-27.9	100.4	-30.8	102.4	-34.6	105.1		
			-19.9	82.4	5.4	1.9	-23.8	83.7	-26.9	84.9	-30.0	86.0	-33.9	87.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	18.5	21.8	34.3	47.6	59.6	70.1	72.9	69.5	62.4	49.6	35.2	23.2
	DBStd	21.67	12.60	12.91	12.69	10.26	9.42	6.88	5.92	5.74	8.52	9.82	11.03	11.64
	HDD50	3875	976	789	504	162	22	0	0	0	9	128	454	831
	HDD65	7234	1441	1209	951	528	222	29	9	22	147	486	894	1296
	CDD50	2849	0	0	18	89	320	602	711	603	379	116	11	0
	CDD65	732	0	0	0	5	55	181	255	160	68	8	0	0
	CDH74	7507	0	0	10	112	697	1927	2582	1393	672	111	3	0
	CDH80	2528	0	0	1	33	254	730	927	384	175	24	0	0
Wind	WSAvg	10.1	10.9	11.1	11.9	13.0	12.3	10.0	7.1	6.5	7.8	9.7	10.6	10.6
Precipitation	PrecAvg	29.9	0.7	0.7	1.7	3.1	3.6	4.6	4.0	3.2	3.1	2.1	1.3	0.8
	PrecMax	41.8	1.4	1.6	3.6	5.8	5.8	9.0	9.7	7.3	9.7	5.5	3.9	2.1
	PrecMin	23.6	0.0	0.0	0.0	0.8	1.3	1.3	0.3	0.6	0.2	0.2	0.0	0.0
	PrecStd	4.4	0.5	0.4	1.0	1.3	1.4	2.0	2.1	1.9	2.1	1.6	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	57.2	74.6	83.9	92.9	94.8	93.4	90.7	89.9	83.7	71.9	54.4
		MCWB	42.2	50.6	57.0	64.1	67.8	74.5	79.7	79.8	71.4	63.9	57.2	49.3
	2%	DB	41.1	49.6	65.6	76.6	86.3	90.7	90.6	87.5	84.5	77.0	63.5	46.6
		MCWB	37.1	45.9	53.6	58.8	65.9	73.3	78.9	77.1	70.6	60.8	52.6	41.4
	5%	DB	36.9	42.7	58.9	71.6	81.6	87.9	88.2	83.9	81.5	71.8	56.9	41.2
		MCWB	34.8	39.1	50.8	55.8	63.8	71.3	77.4	75.9	67.7	58.3	48.0	37.4
	10%	DB	34.4	37.3	52.4	65.5	76.8	84.1	85.7	81.4	77.5	66.1	51.8	37.1
		MCWB	32.9	35.3	47.0	53.2	61.7	69.5	76.0	72.9	65.5	56.9	45.2	34.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	52.5	62.0	65.7	73.5	80.0	84.3	82.1	76.3	69.4	59.9	50.3
		MCDB	47.7	56.2	69.3	81.8	83.9	87.9	89.5	87.0	83.4	75.5	68.5	53.5
	2%	WB	38.0	44.7	56.3	60.7	69.4	77.1	81.0	78.8	73.7	64.2	54.2	42.1
		MCDB	40.7	49.0	63.0	72.4	80.3	86.2	86.7	85.0	80.5	70.7	62.2	46.3
	5%	WB	35.5	39.2	51.8	57.6	66.9	74.8	78.6	76.6	71.5	60.8	49.5	38.0
		MCDB	37.3	42.5	58.8	67.4	77.6	83.8	85.7	82.5	78.4	68.6	56.0	41.3
	10%	WB	33.0	36.0	46.4	54.8	64.3	72.5	76.9	74.3	68.8	57.1	45.3	35.1
		MCDB	34.4	38.0	52.0	63.3	73.0	80.9	83.8	79.9	74.3	64.8	50.5	37.6
Mean Daily Temperature Range	5% DB	MDBR	16.8	16.9	18.3	22.1	22.2	21.6	21.2	20.7	23.7	22.2	19.6	16.2
		MCDBR	19.9	20.1	29.2	31.8	29.4	25.0	22.3	21.5	26.5	29.8	28.9	21.8
		MCWBR	17.1	15.7	18.6	16.9	13.1	11.2	12.8	13.3	13.8	15.8	18.2	17.2
	5% WB	MCDBR	18.8	18.6	27.2	26.0	23.8	21.3	19.8	19.3	21.0	23.8	26.4	20.2
		MCWBR	16.3	14.9	18.3	15.8	12.5	12.3	12.8	13.1	13.0	14.7	17.8	16.4
Clear-Sky Solar Irradiance	taub	0.268	0.282	0.316	0.365	0.388	0.399	0.409	0.406	0.361	0.328	0.291	0.274	
	taud	2.440	2.421	2.418	2.306	2.316	2.346	2.306	2.293	2.396	2.467	2.493	2.449	
	Ebn at Noon	279	293	294	285	281	277	272	269	275	272	268	266	
	Edh at Noon	23	28	32	39	40	39	40	39	33	27	22	21	
All-Sky Solar Radiation	RadAvg	603	947	1222	1527	1773	1978	2096	1741	1399	951	653	484	
	RadStd	57	78	100	133	128	138	89	126	95	117	43	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page